

SLR Parser Visualization

Input: `id ( num + id , id ( num + id ) )`

Original grammar

#	Var	Production
0	C	$\rightarrow \text{id} (A)$
1	A	$\rightarrow A , E$
2	A	$\rightarrow \varepsilon$
3	A	$\rightarrow E$
4	E	$\rightarrow E + T$
5	E	$\rightarrow T$
6	T	$\rightarrow \text{id}$
7	T	$\rightarrow \text{num}$
8	T	$\rightarrow C$

Augmented grammar

#	Var	Production
0	C'	$\rightarrow C$
1	C	$\rightarrow \text{id} (A)$
2	A	$\rightarrow A , E$
3	A	$\rightarrow \varepsilon$
4	A	$\rightarrow E$
5	E	$\rightarrow E + T$
6	E	$\rightarrow T$
7	T	$\rightarrow \text{id}$
8	T	$\rightarrow \text{num}$
9	T	$\rightarrow C$

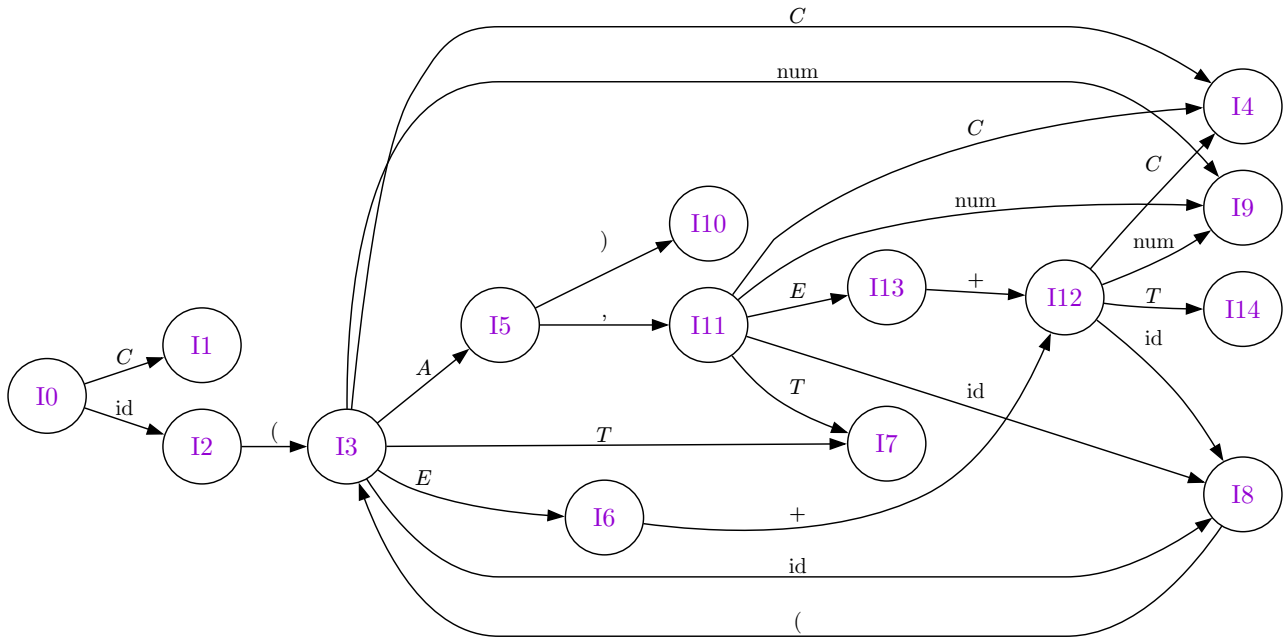
First and Follow sets

Variable	FIRST	FOLLOW
C	id	\$,), ,, +
A	ε , id, num	), ,
E	id, num	), ,, +
T	id, num	), ,, +

Set of items

I₀ <i>inicial</i> $C' \rightarrow \bullet C$ $C \rightarrow \bullet \text{id} (A)$	I₁ (I ₀ , C) $C' \rightarrow C \bullet$	I₂ (I ₀ , id) $C \rightarrow \text{id} \bullet (A)$	I₃ (I ₂ , () $C \rightarrow \text{id} (\bullet A)$ $A \rightarrow \bullet A , E$ $A \rightarrow \bullet \varepsilon$ $A \rightarrow \bullet E$ $E \rightarrow \bullet E + T$ $E \rightarrow \bullet T$ $T \rightarrow \bullet \text{id}$ $T \rightarrow \bullet \text{num}$ $T \rightarrow \bullet C$ $C \rightarrow \bullet \text{id} (A)$	I₄ (I ₃ , C) $T \rightarrow C \bullet$
I₅ (I ₃ , A) $C \rightarrow \text{id} (A \bullet)$ $A \rightarrow A \bullet , E$	I₆ (I ₃ , E) $A \rightarrow E \bullet$ $E \rightarrow E \bullet + T$	I₇ (I ₃ , T) $E \rightarrow T \bullet$	I₈ (I ₃ , id) $T \rightarrow \text{id} \bullet$ $C \rightarrow \text{id} \bullet (A)$	I₉ (I ₃ , num) $T \rightarrow \text{num} \bullet$
I₁₀ (I ₅ ,)) $C \rightarrow \text{id} (A) \bullet$	I₁₁ (I ₅ , ,) $A \rightarrow A , \bullet E$ $E \rightarrow \bullet E + T$ $E \rightarrow \bullet T$ $T \rightarrow \bullet \text{id}$ $T \rightarrow \bullet \text{num}$ $T \rightarrow \bullet C$ $C \rightarrow \bullet \text{id} (A)$	I₁₂ (I ₆ , +) $E \rightarrow E + \bullet T$ $T \rightarrow \bullet \text{id}$ $T \rightarrow \bullet \text{num}$ $T \rightarrow \bullet C$ $C \rightarrow \bullet \text{id} (A)$	I₁₃ (I ₁₁ , E) $A \rightarrow A , E \bullet$ $E \rightarrow E \bullet + T$	I₁₄ (I ₁₂ , T) $E \rightarrow E + T \bullet$

DFA



States table

STATE	ACTION							GOTO			
	id	(	)	,	+	num	\$	C	A	E	T
0	s2							1			
1							acc				
2		s3									
3	s8					s9		4	5	6	7
4			r9	r9	r9						
5			s10	s11							
6			r4	r4	s12						
7			r6	r6	r6						
8		s3	r7	r7	r7						
9			r8	r8	r8						
10			r1	r1	r1		r1				
11	s8					s9		4		13	7
12	s8					s9		4			14
13			r2	r2	s12						
14			r5	r5	r5						

Stack for id (num + id , id (num + id))

Step	Stack	Entry	Action
1	0	id (num + id , id (num + id)) \$	shift 2
2	0 id 2	(num + id , id (num + id)) \$	shift 3
3	0 id 2 (3	num + id , id (num + id)) \$	shift 9
4	0 id 2 (3 num 9	+ id , id (num + id)) \$	reduce T → num
5	0 id 2 (3 T 7	+ id , id (num + id)) \$	reduce E → T
6	0 id 2 (3 E 6	+ id , id (num + id)) \$	shift 12
7	0 id 2 (3 E 6 + 12	id , id (num + id)) \$	shift 8
8	0 id 2 (3 E 6 + 12 id 8	, id (num + id)) \$	reduce T → id
9	0 id 2 (3 E 6 + 12 T 14	, id (num + id)) \$	reduce E → E + T
10	0 id 2 (3 E 6	, id (num + id)) \$	reduce A → E
11	0 id 2 (3 A 5	, id (num + id)) \$	shift 11
12	0 id 2 (3 A 5 , 11	id (num + id)) \$	shift 8
13	0 id 2 (3 A 5 , 11 id 8	(num + id)) \$	shift 3
14	0 id 2 (3 A 5 , 11 id 8 (3	num + id)) \$	shift 9
15	0 id 2 (3 A 5 , 11 id 8 (3 num 9	+ id)) \$	reduce T → num
16	0 id 2 (3 A 5 , 11 id 8 (3 T 7	+ id)) \$	reduce E → T
17	0 id 2 (3 A 5 , 11 id 8 (3 E 6	+ id)) \$	shift 12
18	0 id 2 (3 A 5 , 11 id 8 (3 E 6 + 12	id)) \$	shift 8
19	0 id 2 (3 A 5 , 11 id 8 (3 E 6 + 12 id 8	)) \$	reduce T → id
20	0 id 2 (3 A 5 , 11 id 8 (3 E 6 + 12 T 14	)) \$	reduce E → E + T
21	0 id 2 (3 A 5 , 11 id 8 (3 E 6	)) \$	reduce A → E
22	0 id 2 (3 A 5 , 11 id 8 (3 A 5	)) \$	shift 10
23	0 id 2 (3 A 5 , 11 id 8 (3 A 5) 10	) \$	reduce C → id (A)
24	0 id 2 (3 A 5 , 11 C 4	) \$	reduce T → C
25	0 id 2 (3 A 5 , 11 T 7	) \$	reduce E → T
26	0 id 2 (3 A 5 , 11 E 13	) \$	reduce A → A , E
27	0 id 2 (3 A 5	) \$	shift 10
28	0 id 2 (3 A 5) 10	\$	reduce C → id (A)
29	0 C 1	\$	accept

Parse Tree for `id ( num + id , id ( num + id ) )`

